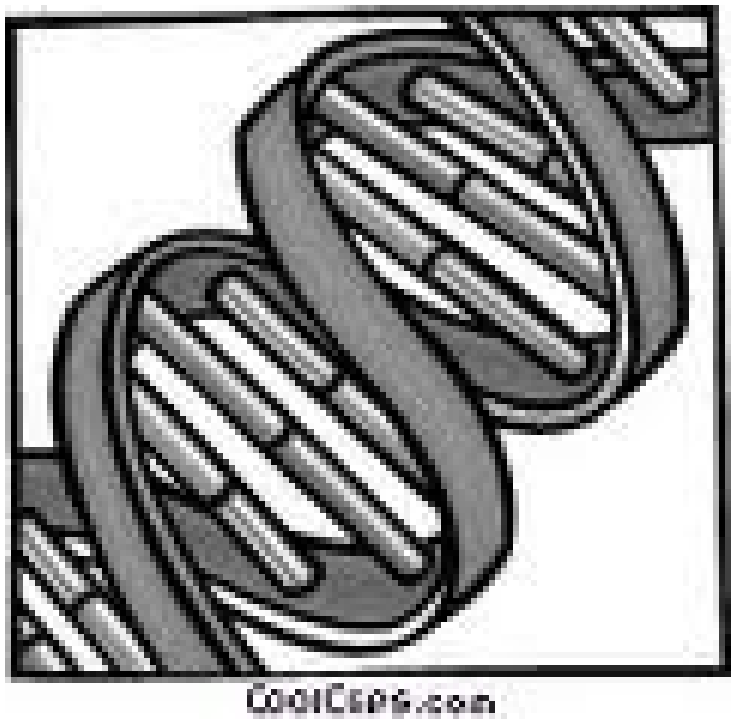
DNA: Structure and Replication



DNA

- DNA: DeoxyriboNucleic Acid
  - Found in nucleus of cells of all living things
  - Contain genetic information
    - Recipe for making proteins used by body and for cell division, replication and repair; and managing chemical reactions (enzymes)
  - Complex molecule arranged in very long strands
  - Human cells have 46 DNA molecules (23 pairs)
    - Each molecule = 1 chromosome.

Learning Aim: Review Structure of DNA

Structure of DNA molecule

- Very LONG strands
- Like a ladder:
  - Sugar phosphate sides
  - Rungs of base pairs
- Base pairs:
  - Joined by hydrogen bonds

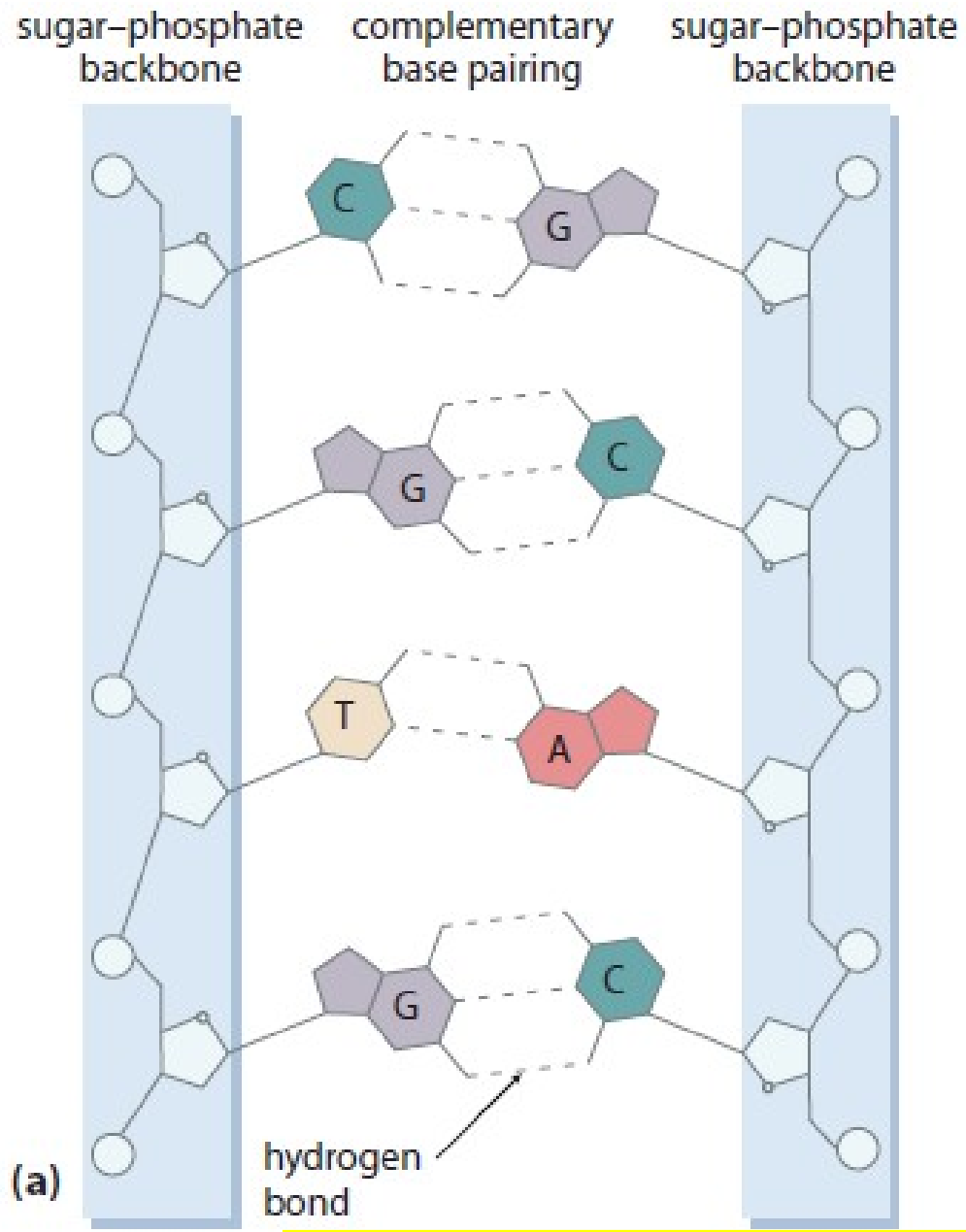
  - C = Cytosine
  - G = Guanine

    C and G always pair

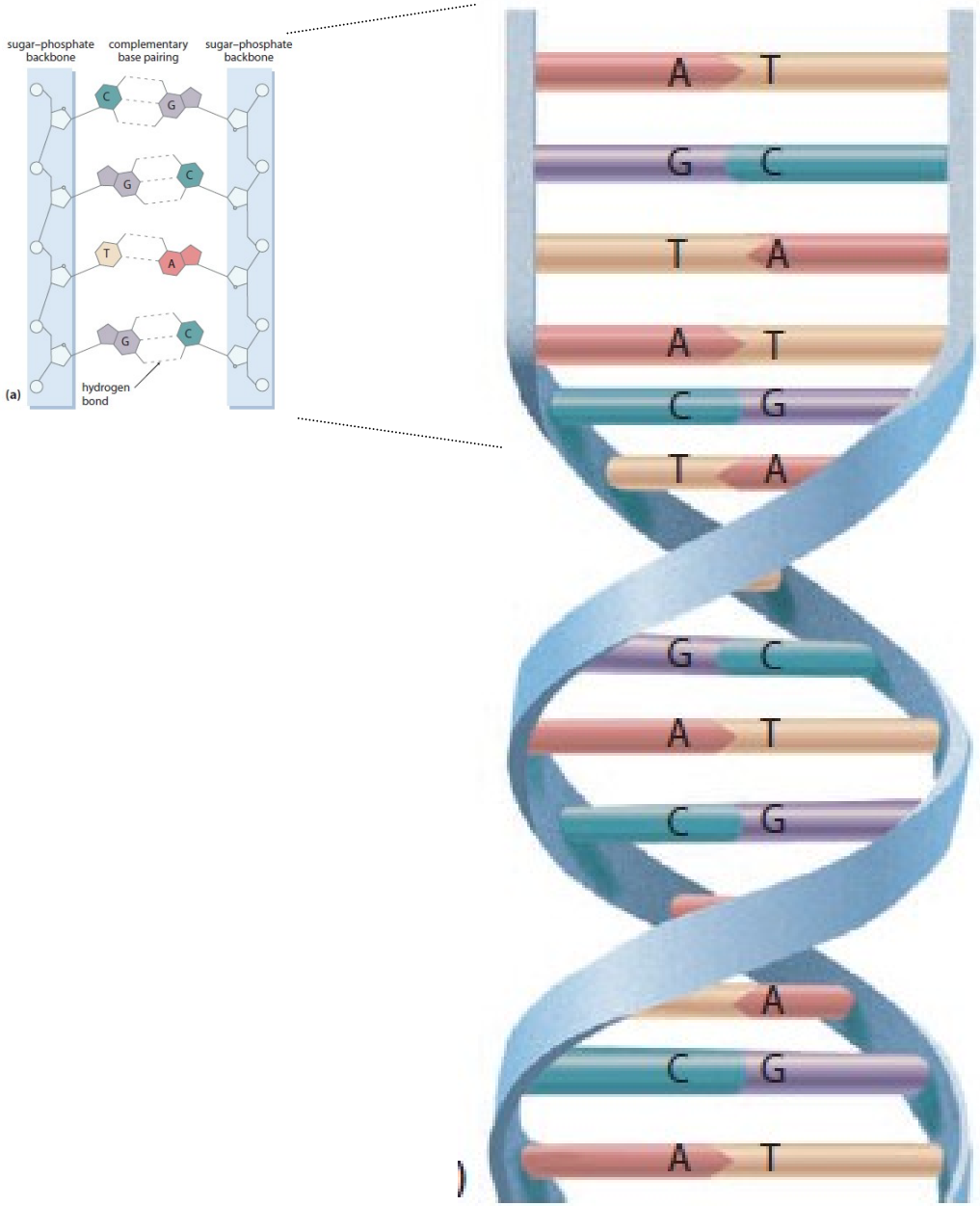
  - T = Thymine
  - A = Adenine

    A and T always pair

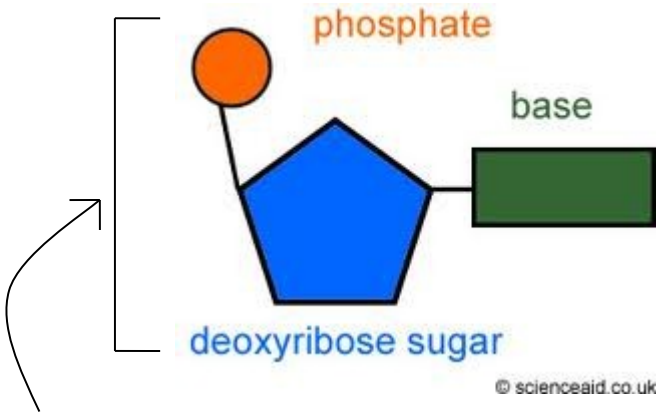
- Strands twisted into helix



Learning Aim: Review Structure of DNA



- Strands are twisted into a "double helix"



Nucleotide = Phosphate, sugar and base

Learning Aim: Review Structure of DNA

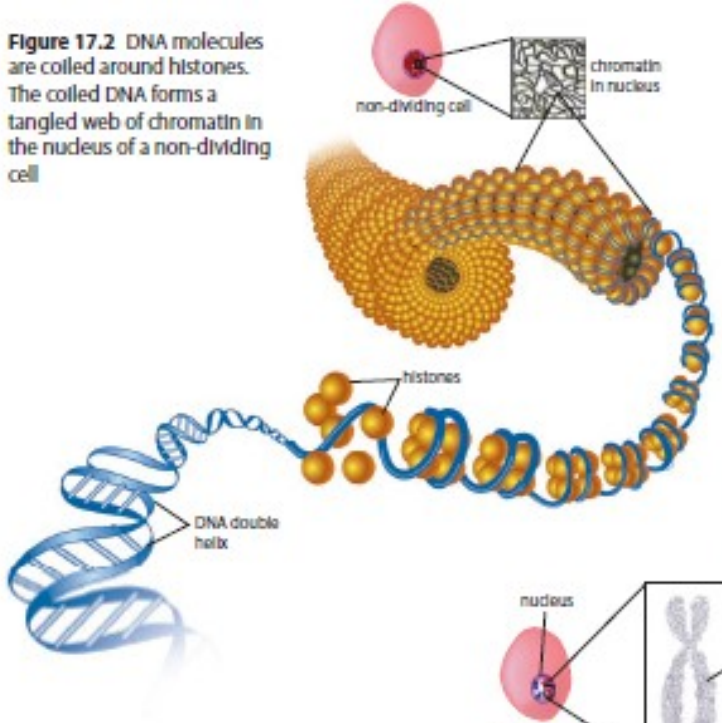


Figure 17.2 DNA molecules are coiled around histones. The coiled DNA forms a tangled web of chromatin in the nucleus of a non-dividing cell

- In a **non-dividing** cell:
  - Double helix wound around histones.
  - Forms coil structure.
  - Forms tangled web of chromatin in nucleus

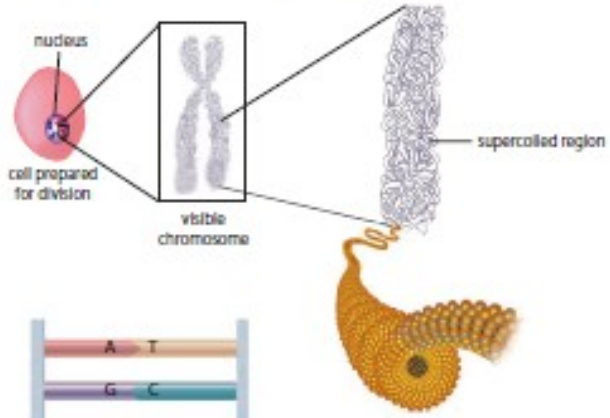
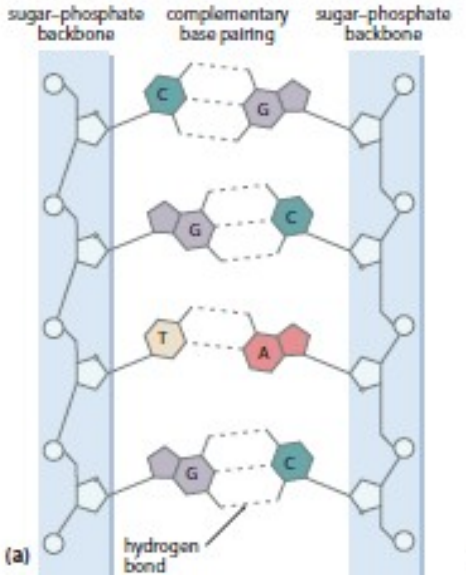


Figure 17.3 For cell division, coils of DNA become super-coiled to form chromosomes

- During **cell division**
  - Coils of DNA become super-coiled
  - Forms chromosome structure



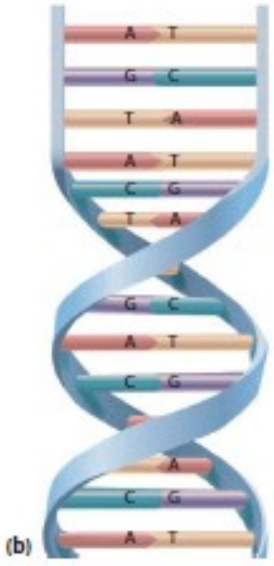
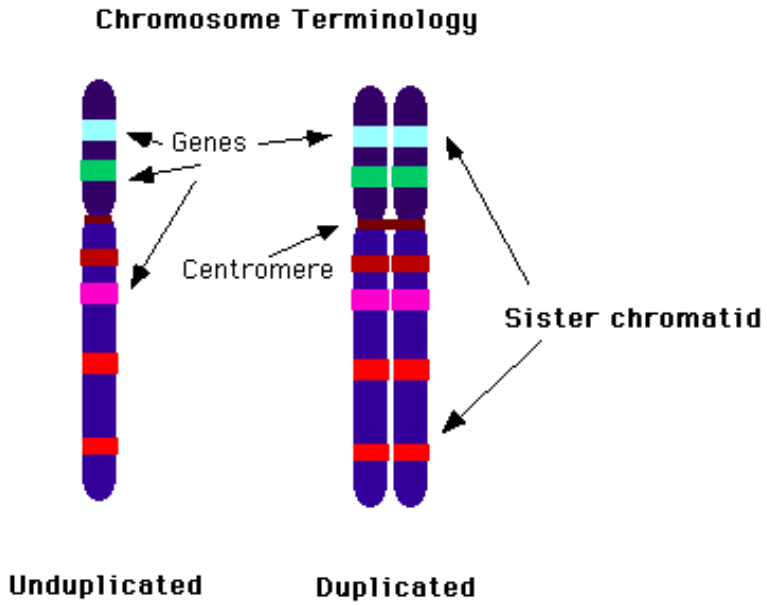


Figure 17.4 (a) DNA consists of two strands of alternating sugars and phosphates with pairs of nitrogen bases forming cross-links between the chains; (b) the two strands are twisted into a double helix

Learning Aim: Review Structure of DNA

DNA Replication

- Replication = copying = duplication
- During interphase, chromatids replicate ready for division:
- Chains with base pairs separate, and new base pairs join to each half.



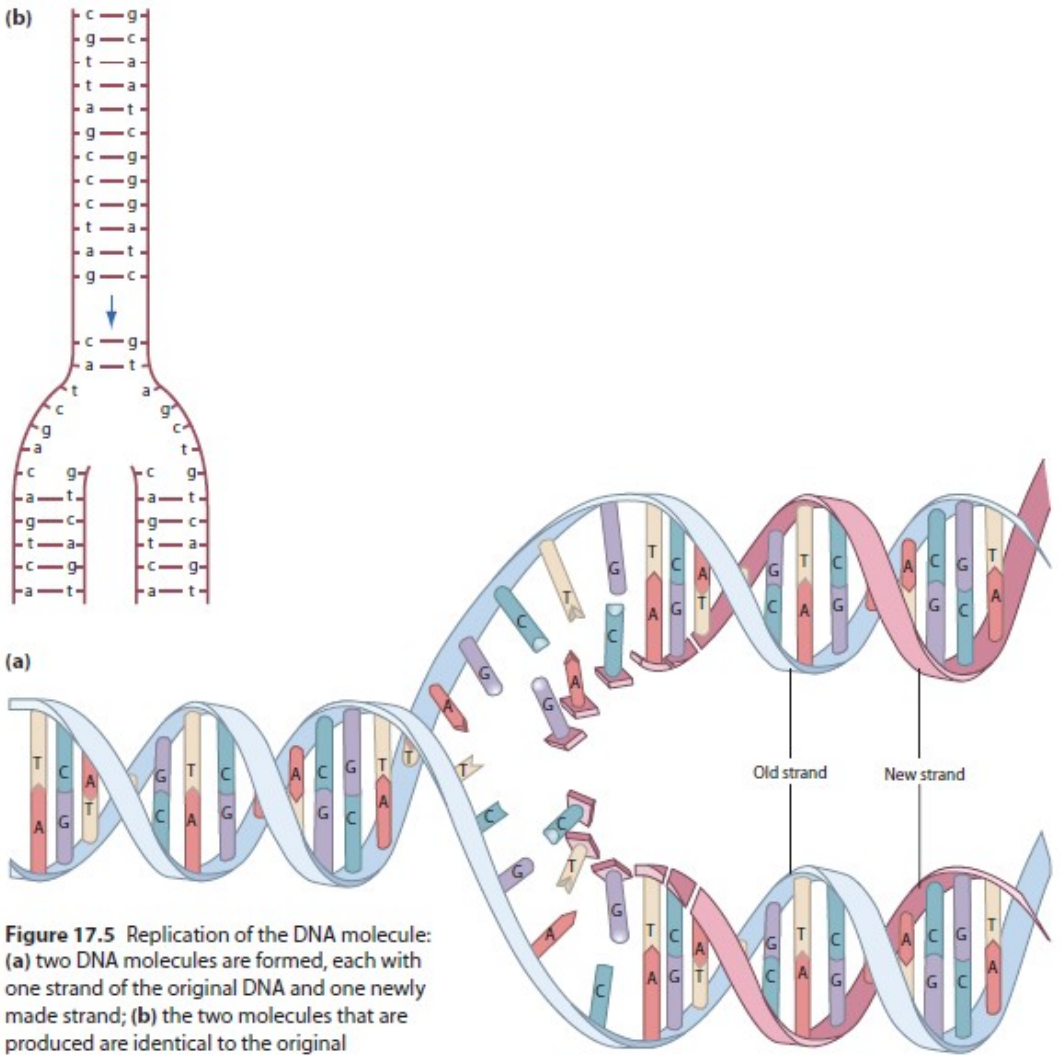


Figure 17.5 Replication of the DNA molecule: **(a)** two DNA molecules are formed, each with one strand of the original DNA and one newly made strand; **(b)** the two molecules that are produced are identical to the original

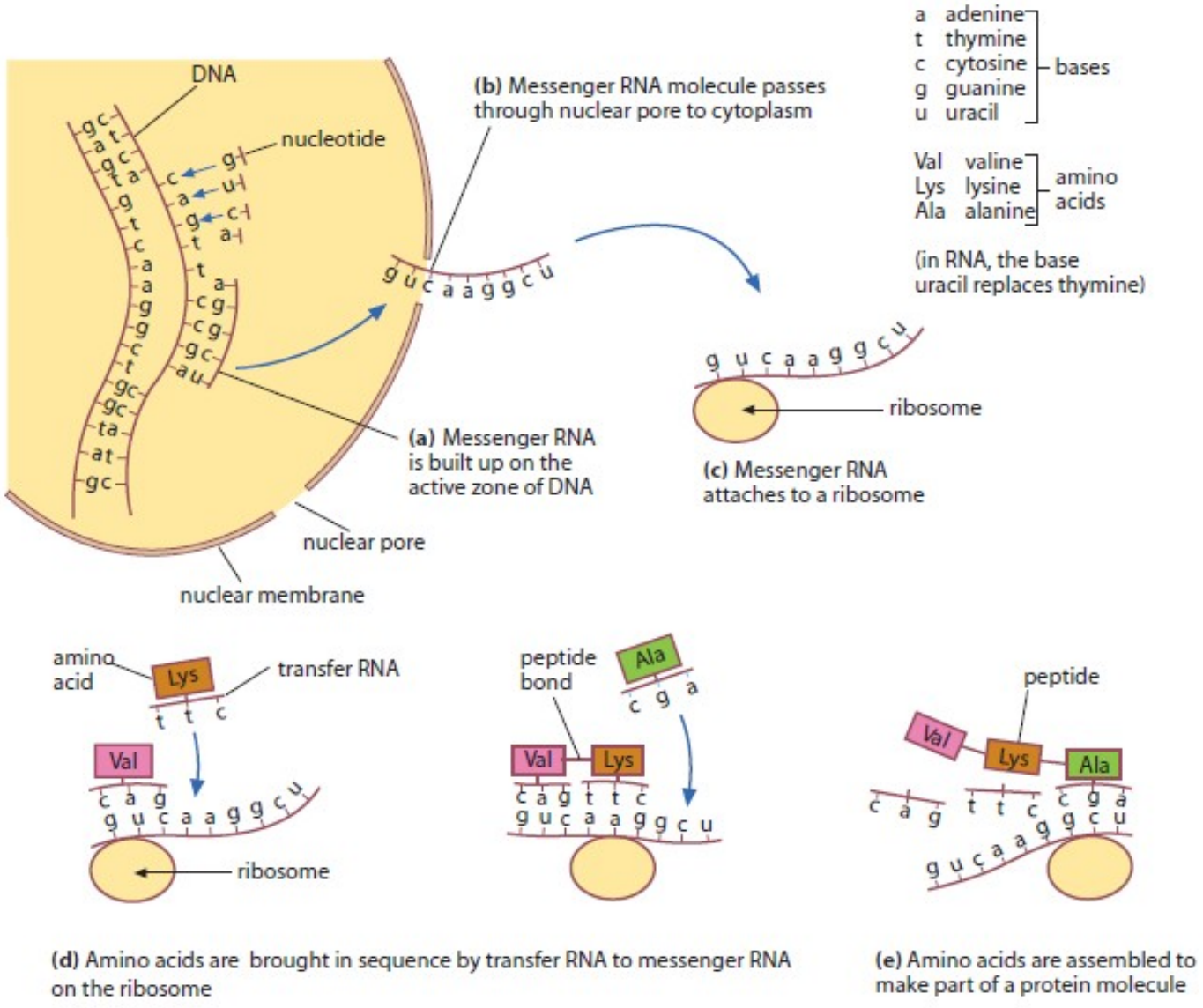
Learning Aim: Review Replication of DNA

DNA makes Proteins



Genes in DNA code for protein production

- Genes are long sequences of DNA base triplets that code for amino acids to be assembled (see picture) into long strands.
- These strands fold, and bind, and form specifically structured proteins.
- Each human gene codes for between 1 and 3 proteins.
- Proteins perform a variety of functions:
  - Structural material for cells
  - Control of chemical reactions (enzymes are proteins)
  - Control of some aspects of membrane transport (as protein channels and/or receptors)
- Genes can be switched on or off depending on the cell's purpose (remember cell differentiation?)

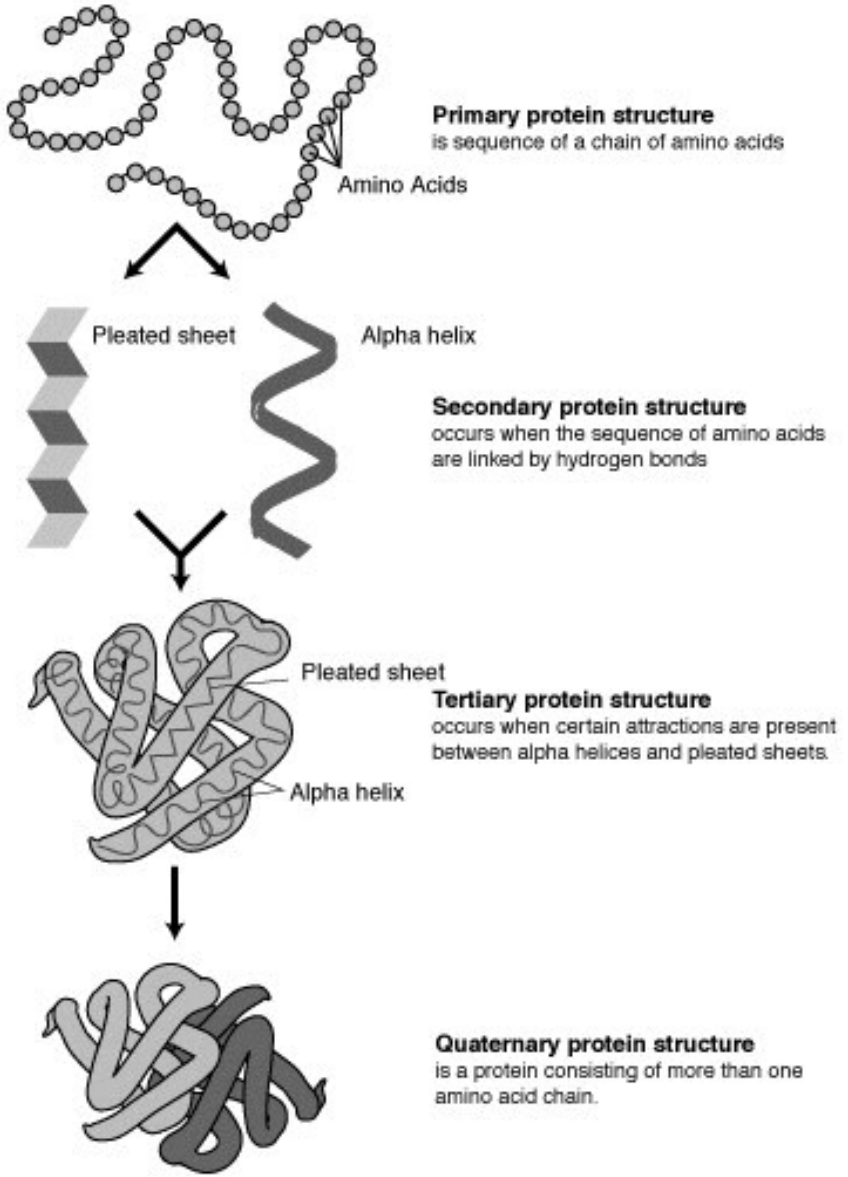


Learning Aim: Review Genes, Protein Synthesis and functions of proteins

Genes in DNA code for protein production

- Proteins are complex structures made from chains of amino acids.
- Lots of different proteins:
  - Different shapes, sizes and functions
  - Made from different combinations of amino acids

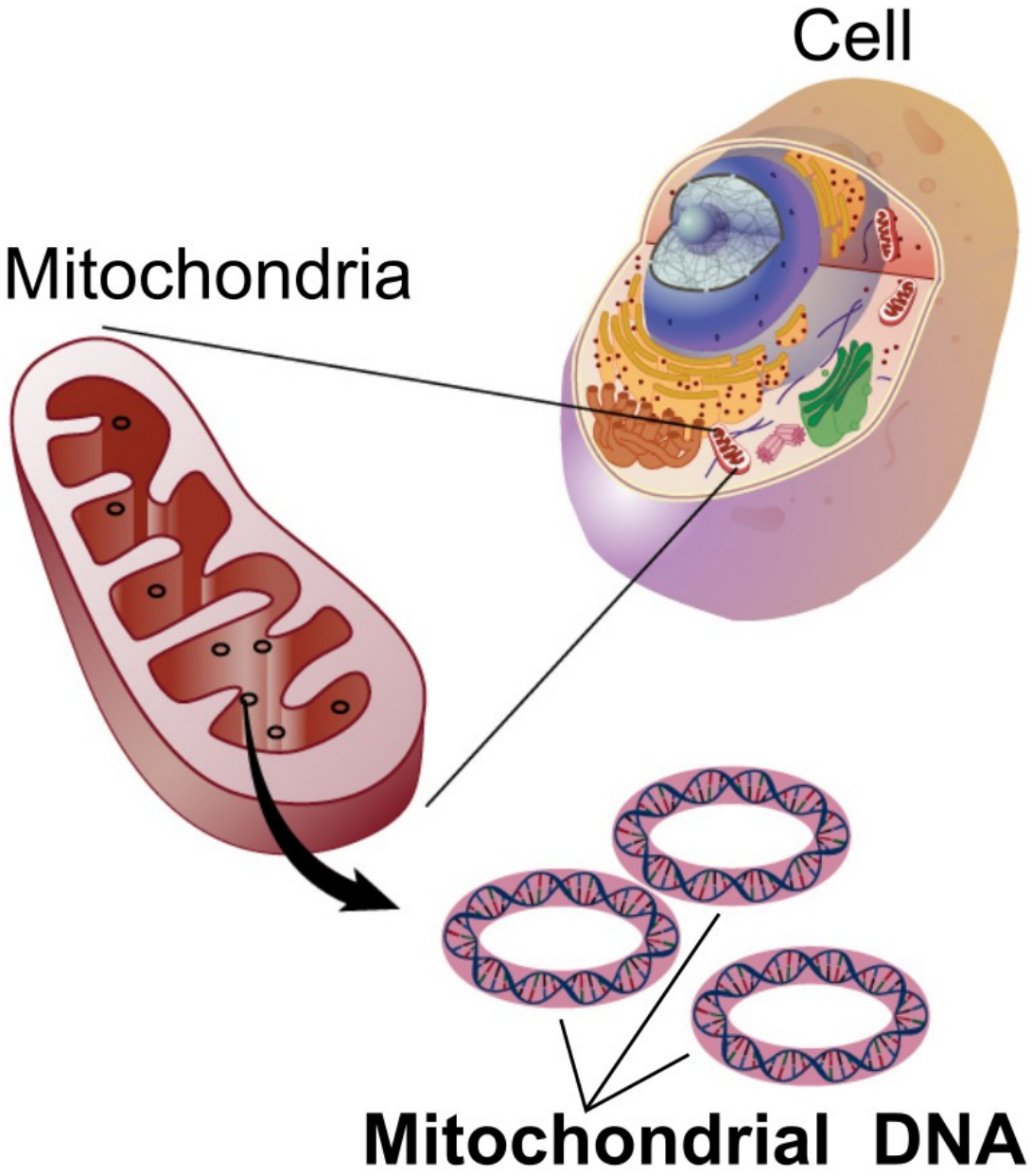
- Assembled by interaction of:
  - Ribosomes
  - tRNA which carries amino acids and a corresponding triplet codon
  - mRNA carrying instructions



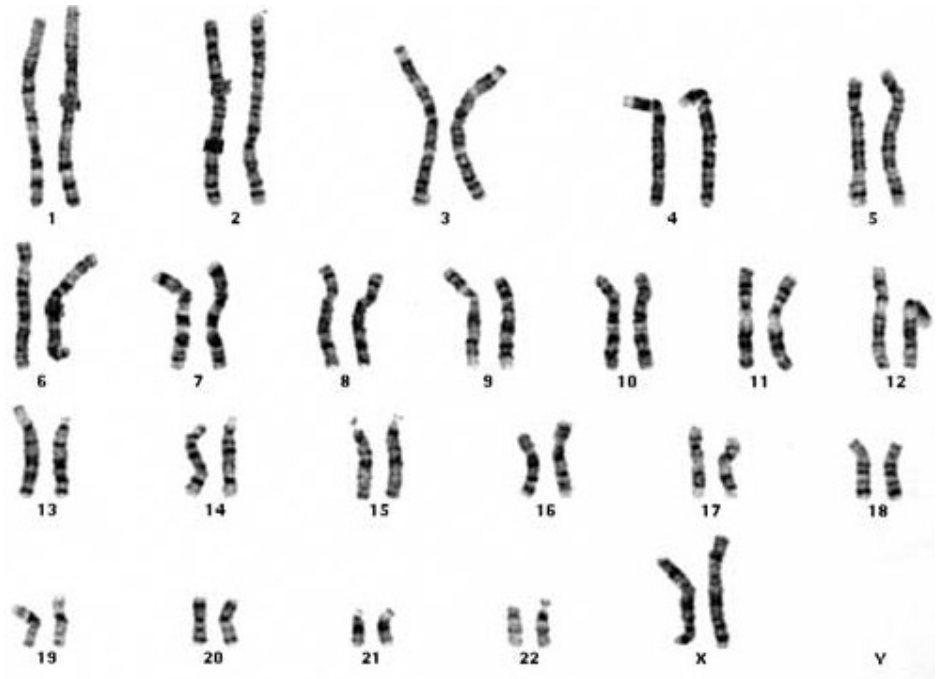
Learning Aim: Review Structure of Proteins

Mitochondrial DNA

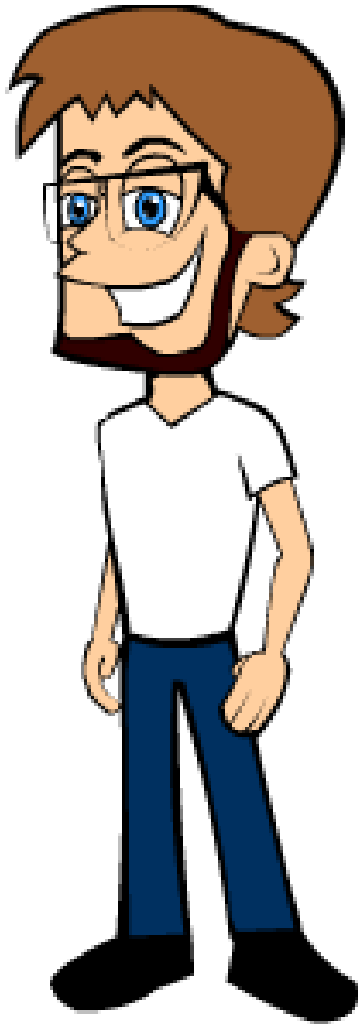
- Mitochondria have their own DNA!
- Small circular molecules rather than large strands
- Codes for enzymes used in cellular respiration
- Inherited entirely through the mother – why?
  - Can trace populations through mitochondrial DNA



DNA and Karyotypes

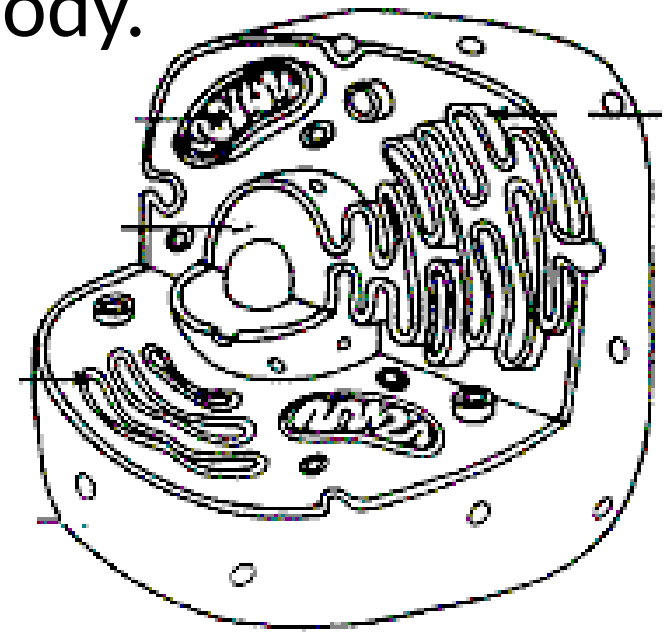


This is Fred.



Learning Aim: Review structure of chromosomes, gene location and alleles

This is one of many cells that make up Fred's body.



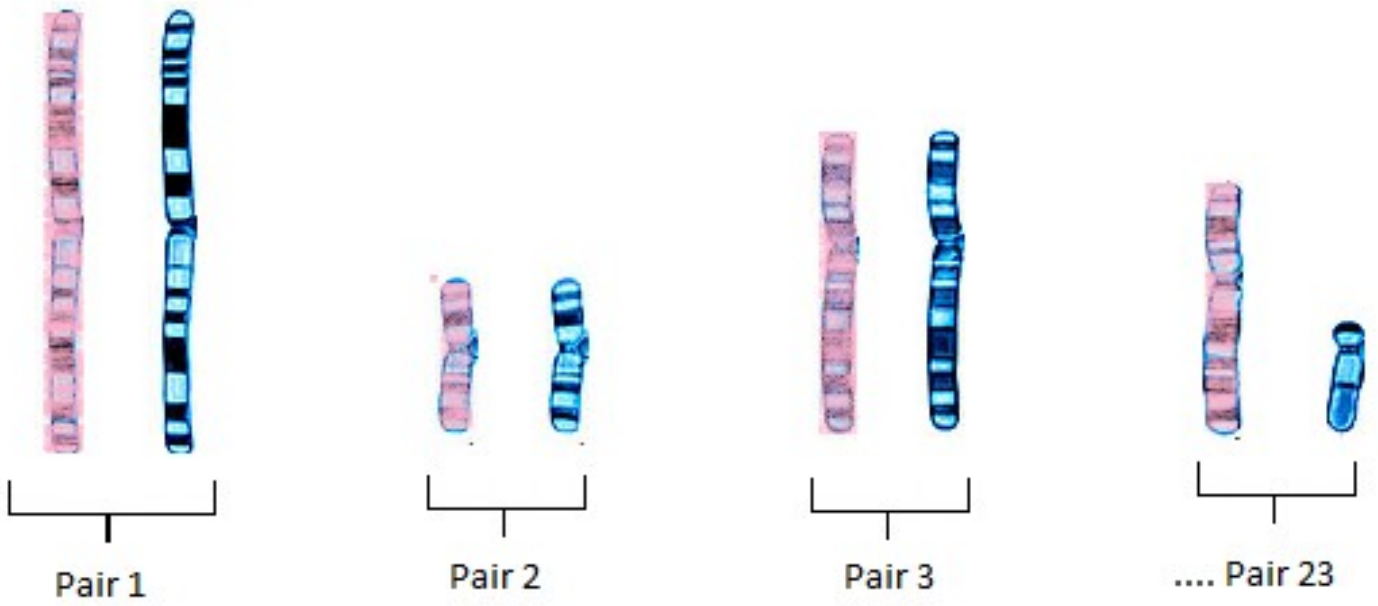
In the nucleus, Fred's cell has DNA, which codes for Fred's characteristics.

Learning Aim: Review structure of chromosomes, gene location and alleles

During Interphase (resting phase), Fred's DNA is arranged in 46 very long strands in the nucleus.

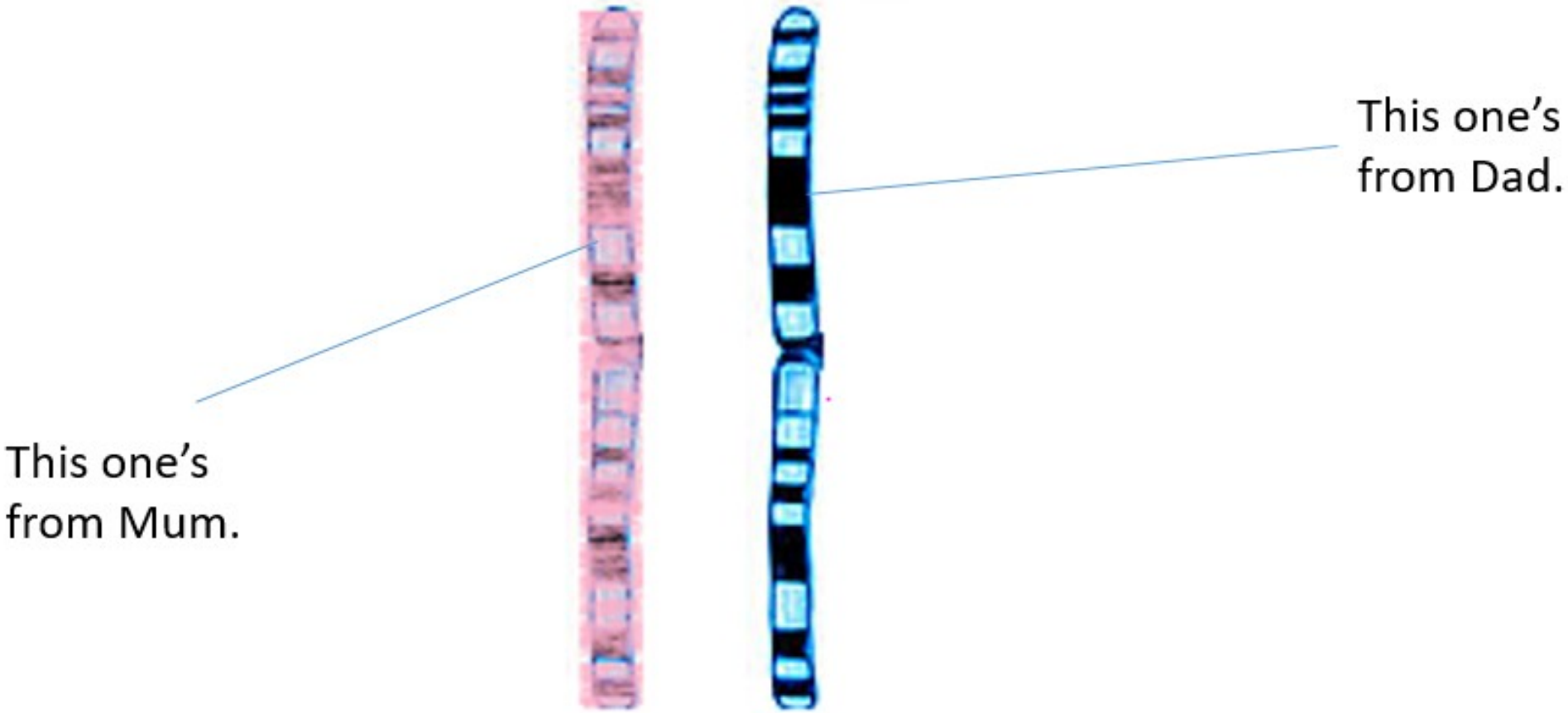
The 46 strands are organised into 23 pairs.

Each pair contains the same genes. There are around 30 000 genes in the human genome (the total number of genes in the 23 chromosome pairs)



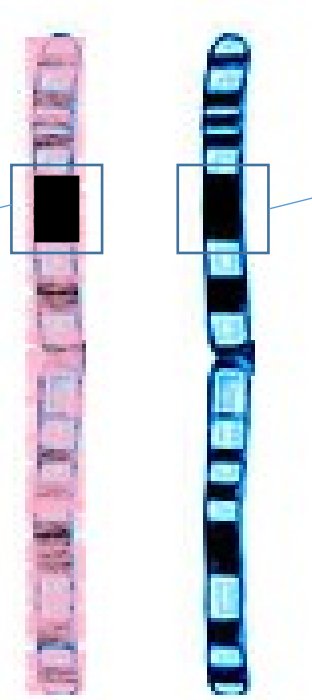
Learning Aim: Review structure of chromosomes, gene location and alleles

Fred has inherited one half of each pair from his Mum, and one from his Dad.



Learning Aim: Review structure of chromosomes, gene location and alleles

Each half of the pair contains genes for the same things, but Fred has inherited different variants (alleles) from each parent:



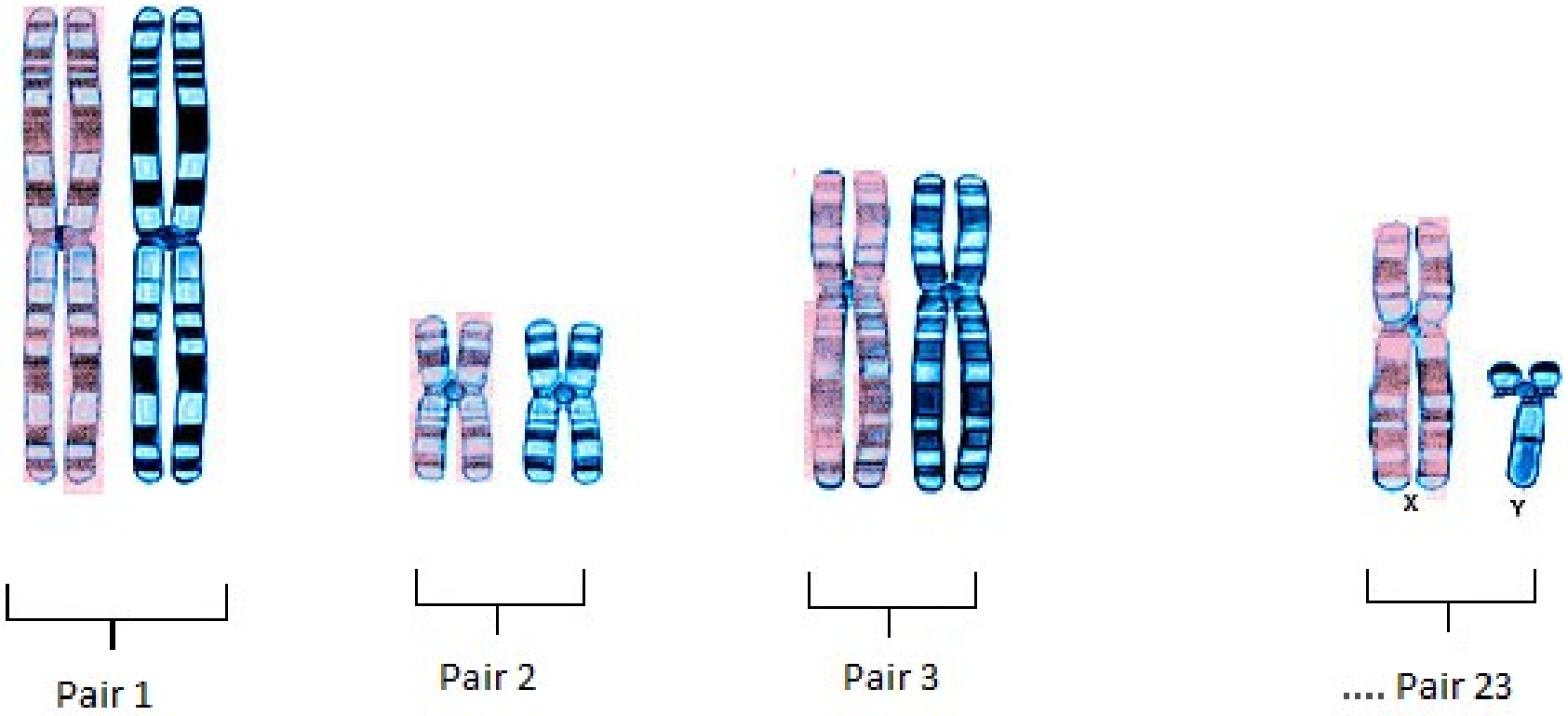
Dimple gene: inherited dominant "dimpling" version from Dad. [D]

Dimple gene: inherited weaker "non-dimpling" version from Mum. [d]

Fred has dimples because the dominant variant (allele) of the gene he got from his dad "overcame" the weaker variant from his Mum. The "dimpling" gene is really a code giving instructions for protein production to the muscles in the cheek, which affects how those cells arrange themselves. One variant of the protein production results in dimples. The other does not. In this case the dimpling gene is dominant.

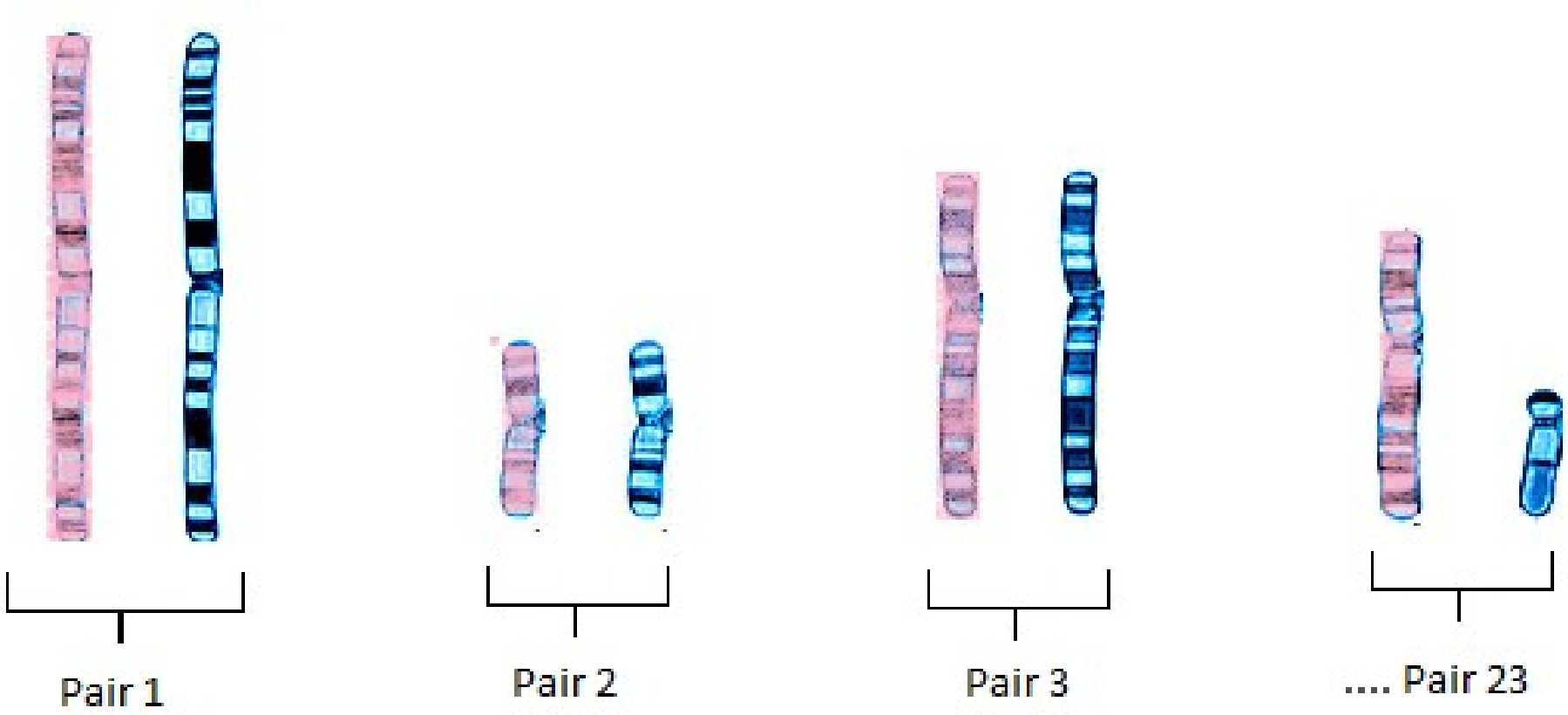
Learning Aim: Review structure of chromosomes, gene location and alleles

When Fred's cell is ready to divide, each of the chromosomes replicates, or copies itself exactly. So now the pairs look like this:



Learning Aim: Review structure of chromosomes, gene location and alleles

When we are discussing DNA and gene inheritance, it is less confusing if we look at the chromosomes during Interphase, like this:



Learning Aim: Review structure of chromosomes, gene location and alleles

- Some traits (eg Fred's dimples) are the result of one gene. Others are the result of several genes on several chromosomes (eg Fred's hair colour)

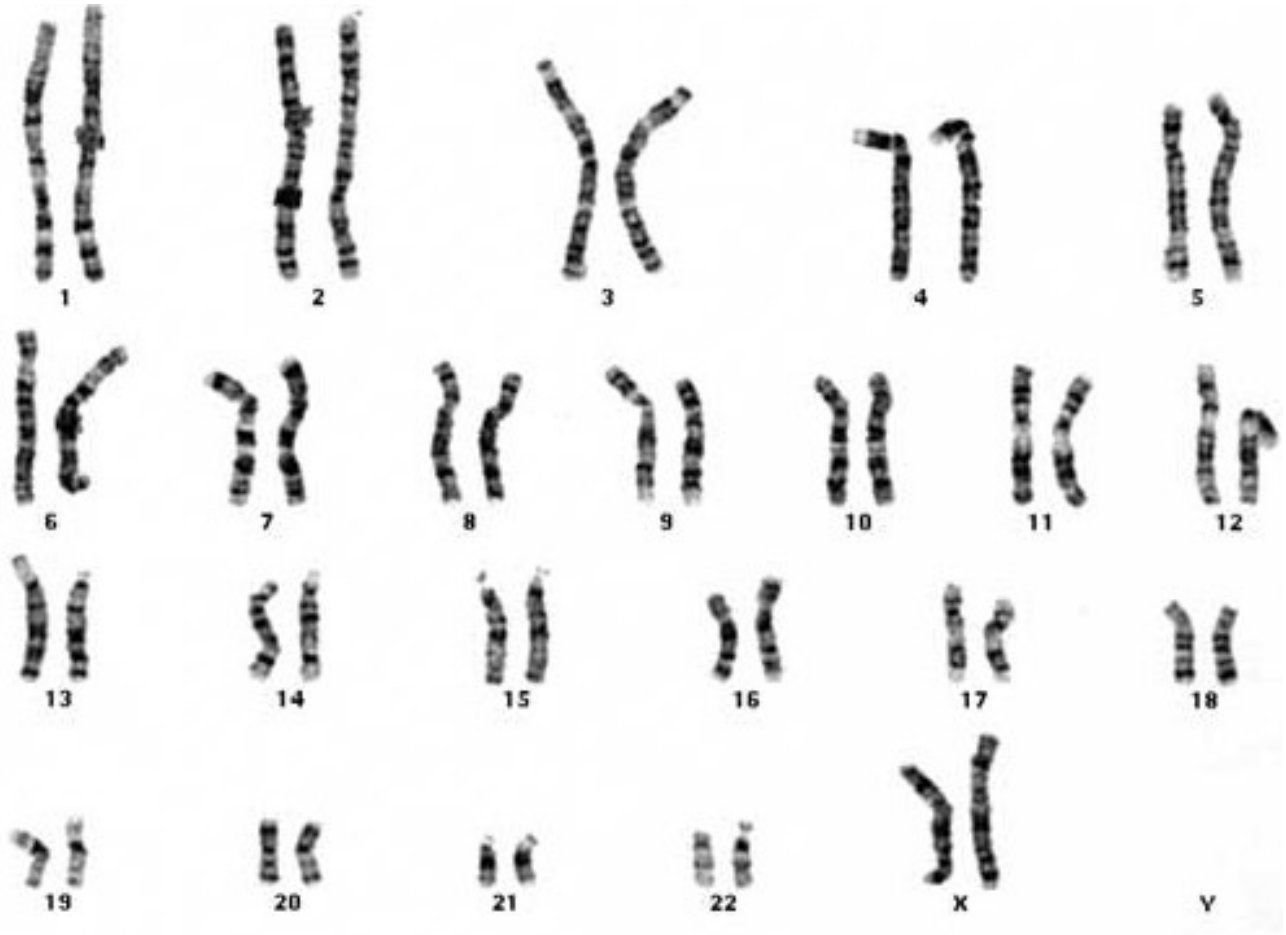
- Some genes have a greater effect on the body than others. A gene controlling whether or not dimples develop doesn't have a huge effect, but inheriting two non-working alleles of the gene that regulates mucus breakdown in the lungs has a big effect: Cystic Fibrosis.

- Another example: inheriting an allele that has mutated to damage tissue can also cause huge problems – eg: Huntington's Disease

Learning Aim: Review effects of genes and alleles

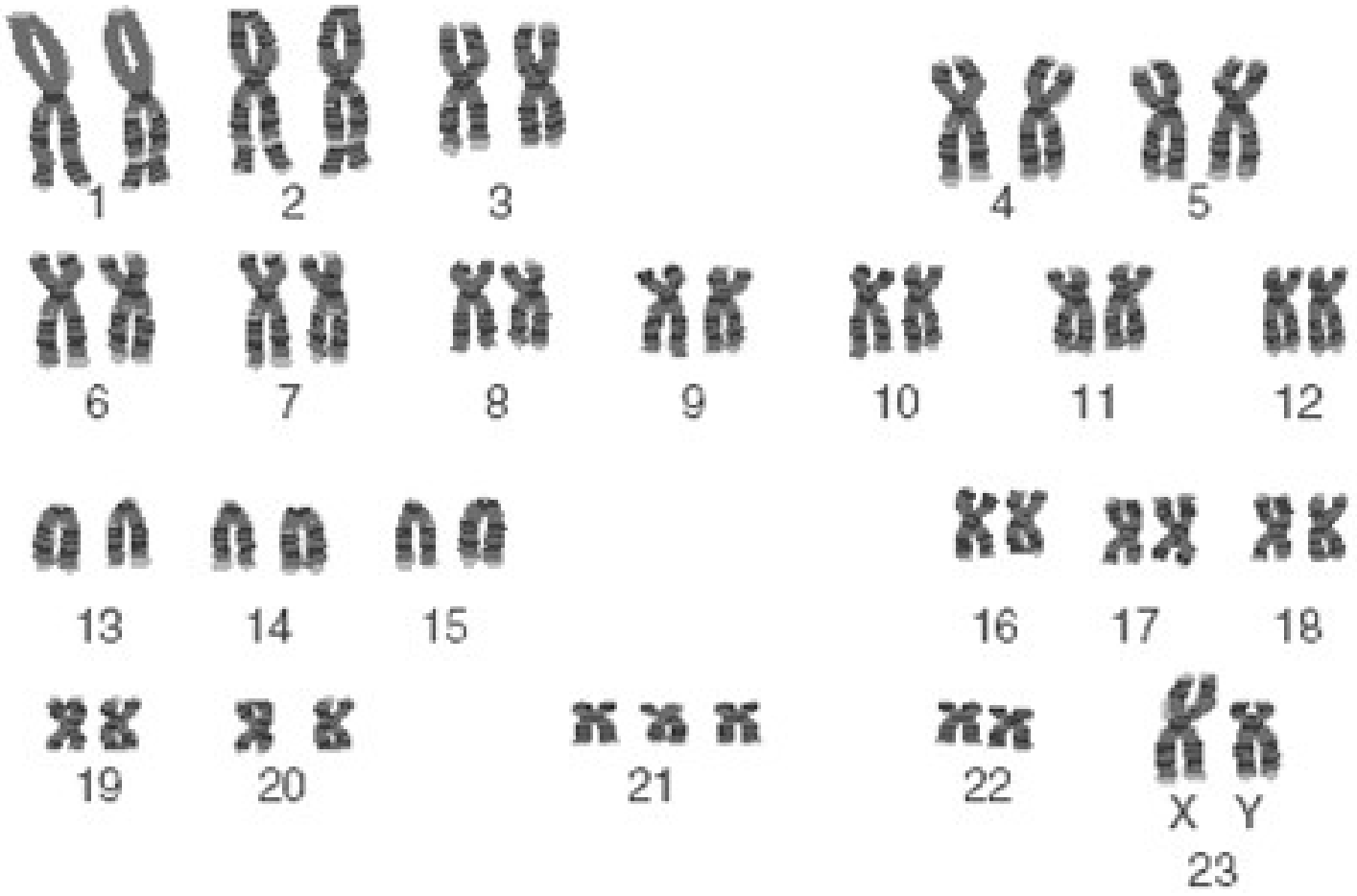
A *karyotype* is a picture of all the chromosomes in a cell.

Sometimes human karyotypes are shown as the chromosomes would appear in interphase:



Learning Aim: Review karyotypes

Other times human karyotypes are shown as the chromosomes would appear just before cell division, when the chromosomes have replicated:



Learning Aim: Review karyotypes